

EE 533: Access Network

Course Code: EE 533

Course Name: Access Network

L-T-P-C: 3-0-2-4

Prerequisites: EE 304 Communication Theory or equivalent, CS 313 Computer Networks or equivalent, or Instructor's consent

Intended for: 3rd/4th year B.Tech. in SCEE; Elective for SCEE PG (M.Tech., M.Tech (R), and Ph.D.)

Mutual Exclusion: None

Approval: 61st BoA (to be reported to Senate)

Preamble

Access networks form the edge of communication infrastructures, connecting users to core networks through wired, optical, and wireless technologies. This course focuses on the Medium Access Control (MAC) layer across broadband, optical, and cellular systems. Students will learn MAC design, resource allocation, QoS provisioning, and energy-efficient operation. The laboratory component, based on OMNeT++/INET and Simu5G, enables hands-on simulation and analysis of MAC protocols in EPON/GPON, WLANs, and LTE/5G NR—equipping students for research and development in next-generation access technologies.

Course Modules

- **Unit 1: Traditional Broadband Access Networks:** Evolution of broadband access; DSL/ADSL/VDSL families and MAC aspects; Ethernet and CSMA/CD; Distributed Queue Dual Bus (DQDB); Token Ring and FDDI. (8 Hours)
- **Unit 2: Optical Access Networks:** Introduction to PONs; EPON/GPON MAC; Dynamic Bandwidth Allocation (DBA); ONU sleep modes; MAC control for energy saving; QoS and service differentiation; trade-offs between latency, throughput, and energy. (11 Hours)
Optional Topics: Advanced Optical Access – TWDM-PON, Long-Reach PON, QoS in DBA, Green PON, ML-based DBA.
- **Unit 3: Wireless Access – WLANs:** Random access mechanisms (ALOHA, Slotted ALOHA, CSMA/CA with RTS/CTS); IEEE 802.11 MAC evolution (DCF, PCF, EDCA, HCF); evolution from Wi-Fi 6/6E (IEEE 802.11ax) to Wi-Fi 7 (IEEE 802.11be) – OFDMA, MU-MIMO, 4K-QAM, Multi-Link Operation (MLO), and punctured transmission; introduction to Wi-Fi 8 (IEEE 802.11bn) features such as Coordinated Spatial Reuse (Co-SR) and Network Performance Control and Adaptation (NPCA), as a preview of emerging research directions. (11 Hours)
Optional Topics: Wireless Sensor Networks – Energy-constrained MACs, duty cycling (S-MAC, T-MAC, X-MAC), IEEE 802.15.4, LoRaWAN, NB-IoT.
- **Unit 4: Cellular Access – LTE & 5G NR:** Resource allocation (RB, RBG, CoReset); Scheduling (Round Robin, Proportional Fair, MAX-C/I, QoS-aware); MAC functions (BSR, DSR, HARQ, logical channel prioritization); Energy efficiency (cDRX, PDCCH monitoring/skipping); C-RAN MAC scheduling and fronthaul constraints. (12 Hours)

Laboratory/practical/tutorial Modules

- Lab 1–2: Introduction to OMNeT++ and INET; environment setup, writing simple modules, running simulations, collecting results.
- Lab 3–4: EPON scheduling experiments; implementing DBA algorithms for TDM/TWDM PON; EPON registration process.
- Lab 5–7: Simulation of random-access MACs (Ethernet CSMA/CD, ALOHA, Slotted ALOHA, CSMA/CA); throughput–delay–efficiency analysis.
- Lab 8–10: Using Simu5G for LTE/5G NR; comparing Round Robin, Proportional Fair, and MAX-CQI schedulers; evaluating XR/video traffic; modifying schedulers for semantic-aware performance.

Text Books

1. D. P. Bertsekas and R. G. Gallager, *Data Networks*, 2nd ed., Prentice Hall, USA, 1992.
2. G. Kramer, *Ethernet Passive Optical Networks*, McGraw-Hill, 2005.

References

1. S. Sesia, I. Toufik, and M. Baker, *LTE: The UMTS Long Term Evolution – From Theory to Practice*, 2nd ed., Wiley, UK, 2011.
2. G. Kramer, B. Mukherjee, and G. Pesavento, “IPACT: A dynamic protocol for an Ethernet PON (EPON),” *IEEE Commun. Mag.*, vol. 40, no. 2, pp. 74–80, 2002.
3. M. P. McGarry, M. Reisslein, and M. Maier, “Ethernet passive optical network architectures and dynamic bandwidth allocation algorithms,” *IEEE Communications Surveys & Tutorials*, vol. 10, no. 3, pp. 46–60.
4. K. Kanonakis and I. Tomkos, “Improving the efficiency of online upstream scheduling and wavelength assignment in hybrid WDM/TDMA EPON networks,” *IEEE J. Sel. Areas Commun.*, vol. 28, no. 6, pp. 838–848, Aug. 2010.
5. W. Ye, J. Heidemann, and D. Estrin, “An energy-efficient MAC protocol for wireless sensor networks (S-MAC),” in *Proc. IEEE INFOCOM*, 2002, pp. 1567–1576.
6. T. van Dam and K. Langendoen, “An adaptive energy-efficient MAC protocol for wireless sensor networks (T-MAC),” in *Proc. ACM SenSys*, 2003, pp. 171–180.
7. *IEEE Standard for Service Interoperability in Ethernet Passive Optical Networks (SIEPON)*, IEEE Standard 1904.1, 2017.
8. IEEE Standard 802.11 (WLAN MAC/PHY) and IEEE 802.3ah (Ethernet in the First Mile). 3GPP, “NR; Physical channels and modulation (Release 17),” 3GPP TS 38.211, v17.x.x, Sep. 2023.